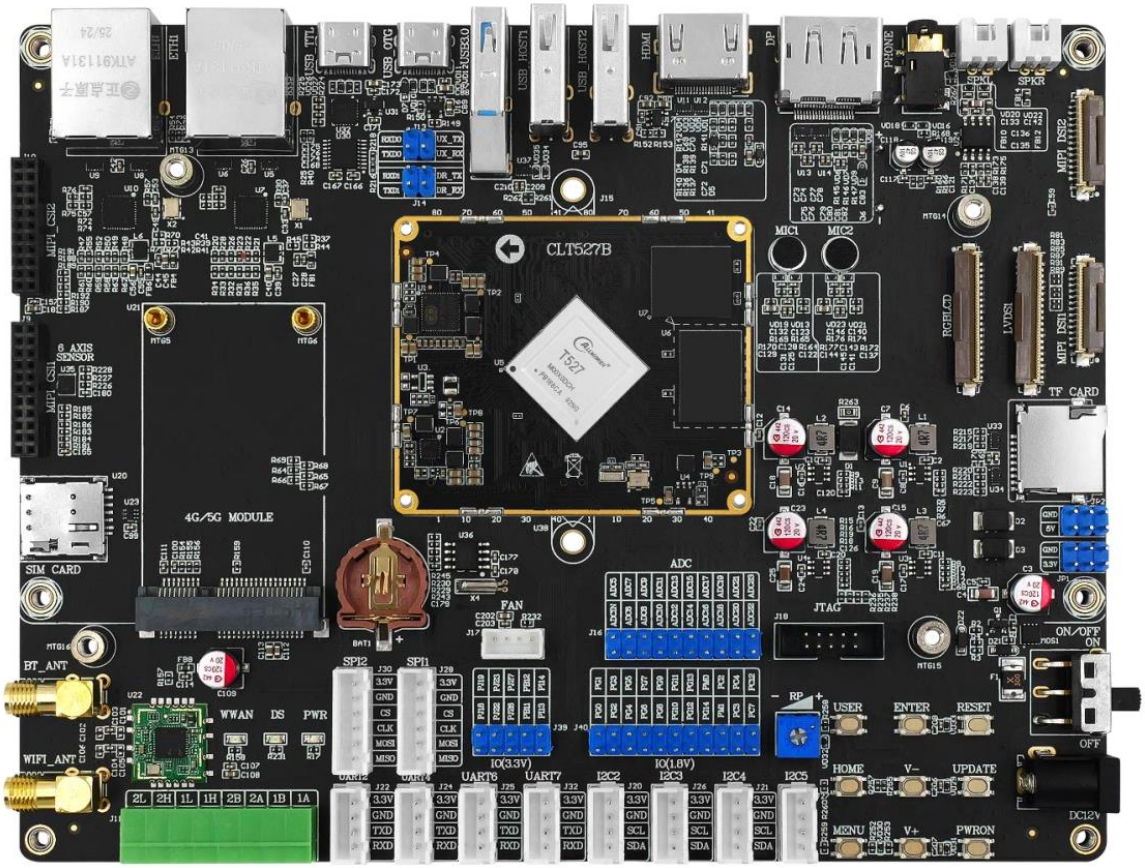
ATK-DLT527B

Development Board Specification

V1.0



ALIENTEK

1. Shopping:

TMALL: https://zhengdianyuanzi.tmall.com

TAOBAO: https://openedv.taobao.com

2. Download

Address: http://www.openedv.com/docs/index.html

3. FAE

Website : www.alientek.com

Forum : http://www.openedv.com/forum.php

Videos : www.yuanzige.com

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In order to get the latest version of product information, please regularly visit the download center or contact the customer service of Taobao ALIENTEK flagship store. Thank you for your tolerance and support.

Revision History:

Version	Version Update Notes	Responsible person	Proofreading	Date
V1.0	release officially	ALIENTEK Linux Team	ALIENTEK Linux Team	2025.03.01

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Chapter 1. Overview of Development Board

1.1 Introduction to the development board

The ALIENTEK ATK-DLT527B development board is a high-performance embedded development platform based on the domestic Allwinner T527 octa-core industrial-grade processor, providing abundant hardware interfaces and complete software and hardware support. The development board consists of the ATK-CLT527B core board and a baseboard, connected via board-to-board (BTB) connectors. The core board is equipped with LPDDR4X high-speed memory and large-capacity EMMC storage as standard, meeting the storage capacity requirements of most products. Customized memory and storage configurations are also available according to bulk demand.

The ALIENTEK ATK-CLT527B core board adopts the Allwinner T527 octa-core industrial-grade high-performance processor. As a new-generation, high-performance and low-cost AI platform solution launched by Allwinner for the smart device market, the T527 is mainly targeted at the mid-to-high-end smart device segment.

The T527 chip series integrates multiple high-performance computing units, including an 8-core 64-bit Cortex-A55 CPU, G57 MC1 GPU, HiFi4 DSP, RISC-V MCU, and a 2 TOPS NPU (NPU is only supported by the T527M02X0DCH model, which is not used in the current ATK-CLT527B core boards sold by our company). The CPU operates at a maximum frequency of 1.8GHz.

It features powerful hardware encoding and decoding capabilities, rich display and camera interfaces, and supports common interfaces such as CAN, PCIe, SDIO, USB, GMAC, SPI, TWI, UART, GPADC, LRADC, and PWM. It can be widely used in content sharing and self-service interactive terminals, industrial intelligence, industrial control, automotive electronics, intelligent manufacturing, and other commercial and industrial electronic equipment.

At the same time, it supports operating systems including Linux and Android to meet the requirements of complex applications.

The baseboard of the development board is equipped with **rich peripheral resources**, including 2× Gigabit Ethernet, 2× RS485, 2× CAN, 1× USB2.0 OTG, 1× USB3.0 HOST, 2× USB2.0 HOST, 1× RGB display interface, 1× LVDS display interface, 2× MIPI display interfaces, 1× HDMI output, 1× DP output, 2× MIPI camera interfaces, 4× TTL UARTs, 4× I2C interfaces, 2× SPI interfaces, 1× 4G/5G module interface, onboard WiFi & BT, and audio functions. These facilitate users in early project requirement evaluation, functional testing and verification, and accelerate project development.

ALIENTEK provides abundant software resources for the **ATK-DLT527B** development board, including but not limited to Tina5.0 Linux SDK, Android13 SDK, Linux Buildroot & Android13 firmware, Qt source code, relevant development tools and environments. Corresponding development documents are also provided, covering development environment setup, SDK compilation, packaging & burning instructions, hardware testing guides, hardware design references, etc.

In addition, ALIENTEK has specially compiled a series of materials for core board users for the development phase, including schematics, baseboard design references, mechanical structures, component footprints, connector specifications, Linux Buildroot & Android firmware and source code, compilers, software packages, etc., enabling users to achieve rapid development, improve efficiency and shorten development cycles.

ALIENTEK development boards are supported by **complete and practical documentation**, helping customers shorten development cycles and accelerate time-to-market.

1.2 Typical Application Areas

It can be widely used in content sharing and self-service interactive terminals, industrial intelligence, industrial control, automotive electronics, intelligent manufacturing, and other commercial and industrial electronic equipment fields.

Chapter 2. Development Board Resources

2.1 Hardware Resources

The baseboard of the ATK-DLT527B development board is equipped with abundant peripheral interfaces, maximizing the internal resources of the T527 processor. Most functions of the main controller can be verified on this development board. Meanwhile, a wealth of interfaces and functional modules are expanded to facilitate users in early project requirement evaluation, functional testing and verification.

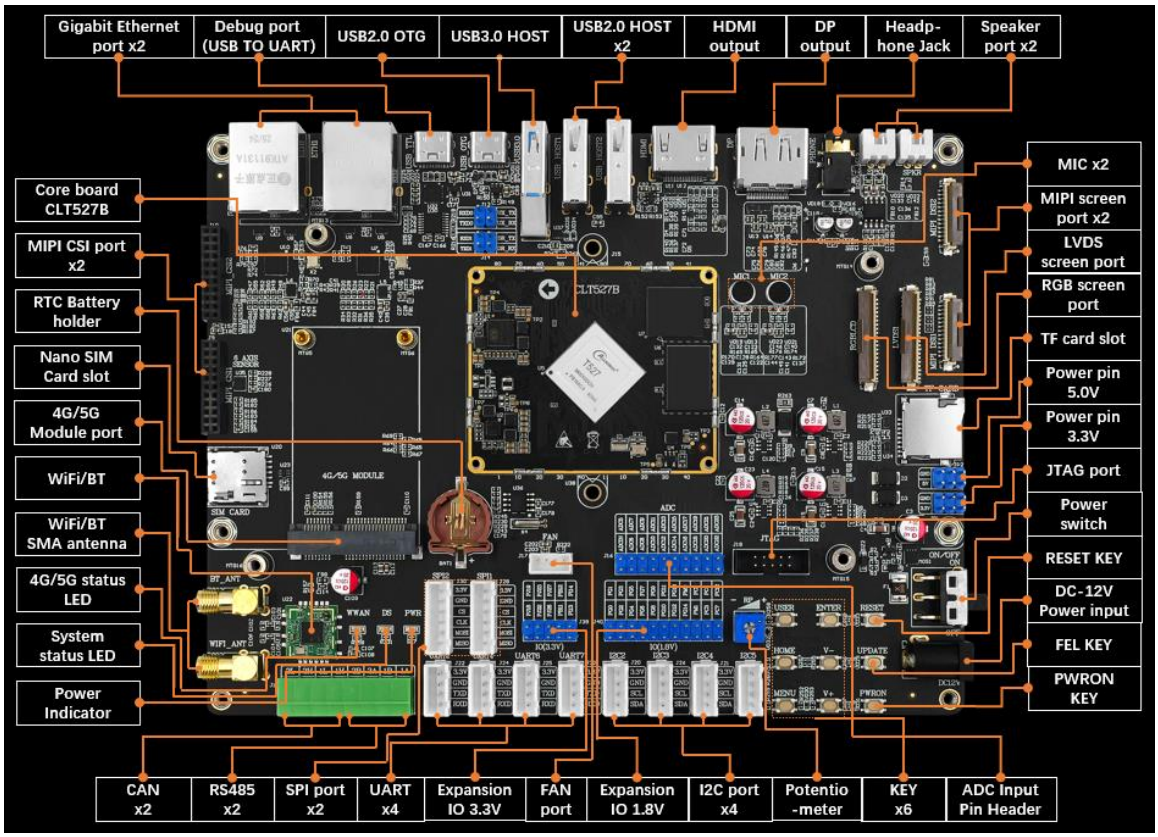


Figure 2.1-1 Hardware Resource Block Diagram of the Development Board

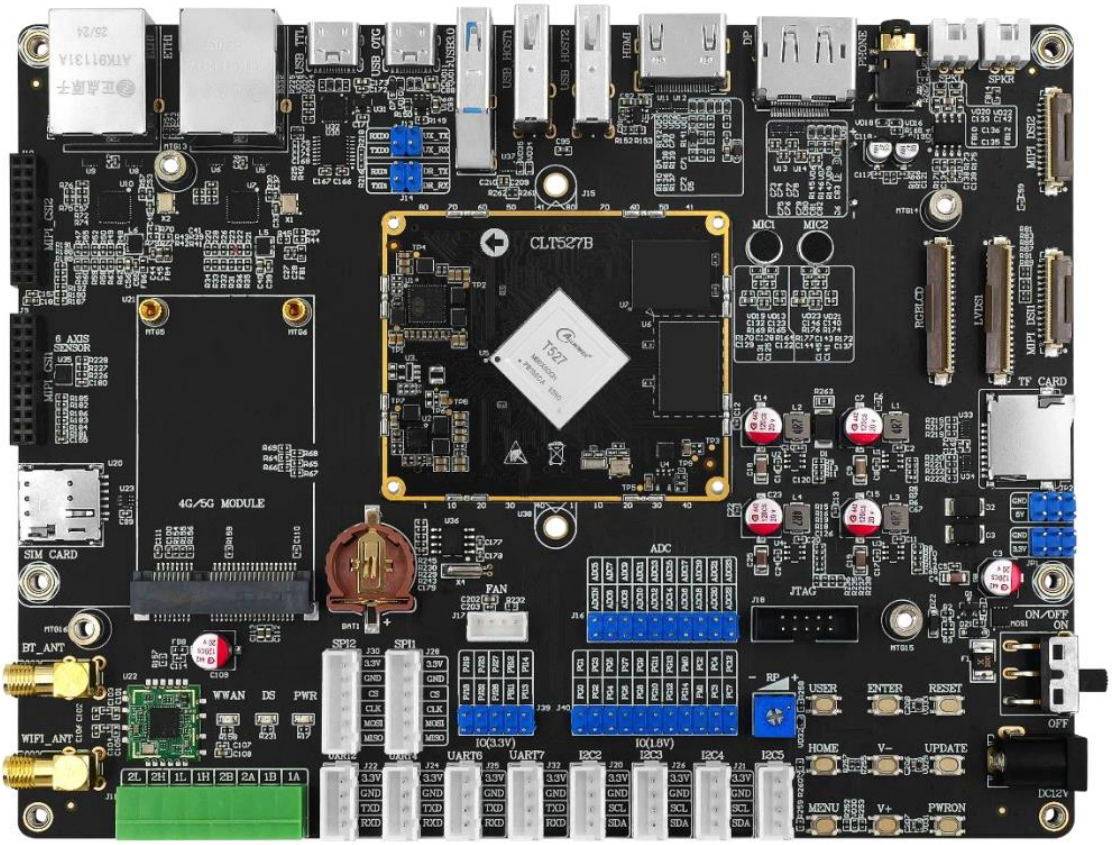


Figure 2.1-1 Front view of the development board

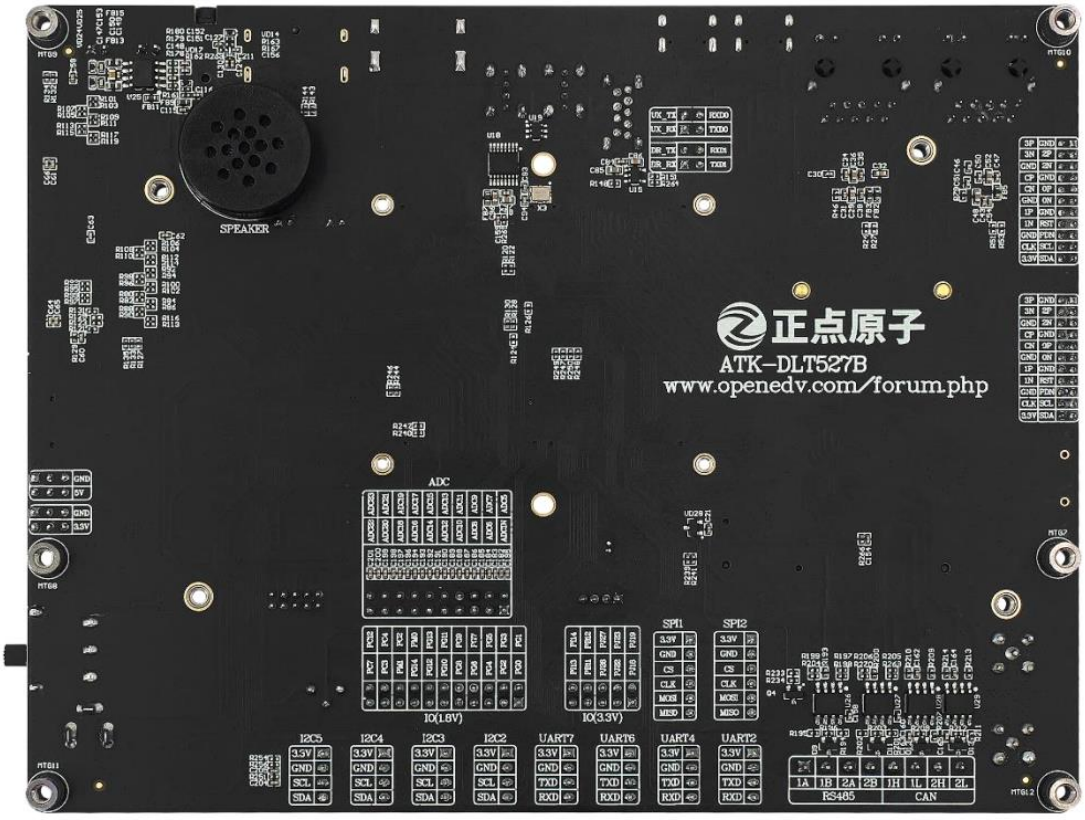


Figure 2.1-2 Back view of the development board

The features of the ALIENTEK ATK-DLT527B development board include:

1. Rich interfaces. The development board provides various standard interfaces, facilitating the experiment and development of various peripherals.

2. Flexible design. Adopting a core board + baseboard structure, many peripheral interfaces on the board can be flexibly configured to meet usage requirements under different conditions. The core board leads out almost all communication interfaces of the T527 processor, including UART, I2C, I2S, SPI, CAN, SDMMC, LVDS/MIPI DSI, MIPI CSI, USB 2.0, USB 3.0/PCIE 2.1, ADC, Gigabit Ethernet, etc., which greatly facilitates user expansion and application.

3. User-friendly design. Each interface is clearly labeled with silkscreen printing for easy identification; commonly used peripherals are marked with large silkscreen for quick location; the interface positions are reasonably arranged for convenient operation.

2.2 Resource Parameters

The block diagram of T527 processor parameters is shown in the figure below.

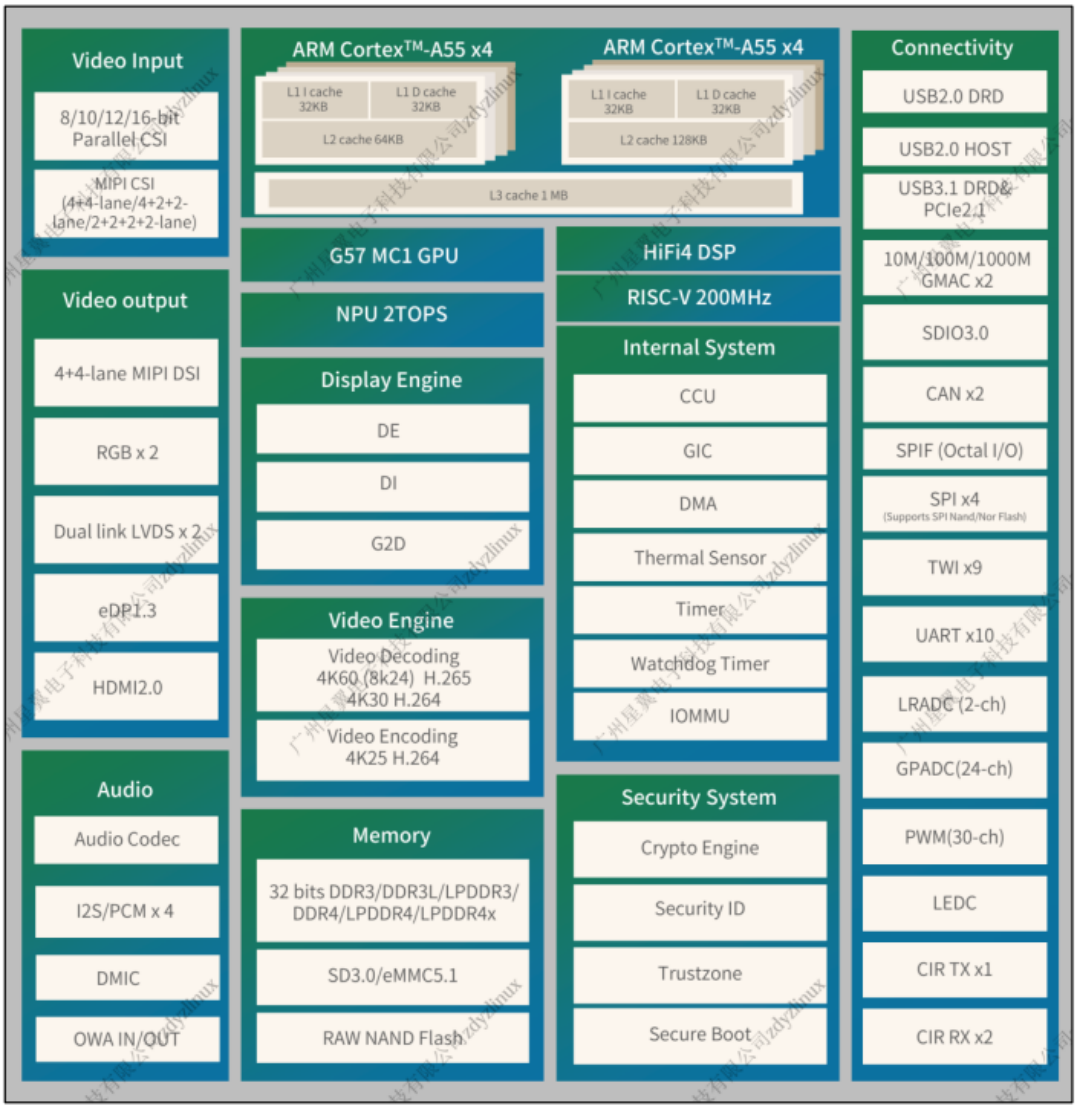


Figure 2.2-1 T527 Processor Block Diagram

The development board consists of a core board and a baseboard connected via BTB connectors.

The resource parameters of the core board are shown in the table below.

Table 1 Core Board Resource Parameters

ATK-CLT527B Core Board Resource Parameters	
CPU	8-core 64-bit ARM Cortex-A55, main frequency up to 1.8 GHz
GPU	ARM Mali-G57 Support OpenGL ES 3.2/2.0/1.1, Vulkan 1.1/1.2/1.3, and OpenCL 2.2
MCU	RISC-V XuanTie E906, main frequency 200 MHz
DSP	Cadence Tensilica HiFi4 DSP
G2D	- 2D image acceleration engine, maximum resolution: 2560×2560 - Supports image rotation, mirroring, cropping and stitching, image data downsampling and stepless scaling, data format conversion, color space conversion, etc.
Video Encoding	· H.264 - 4K@25fps · MJPEG - 4K@15fps · JPEG - 8192*8192
Video Decoding	· H.265 - 4K@60fps · VP9 - 4K@60fps · H.264 - 4K@30fps · H.263/VP8/AVS/AVS+/MPEG-1/MPEG-2/MPEG-4 - 1080P@60fps · JPEG - 16384*16384
Video Input	· Parallel CSI - 8/10/12/16-bit width · MIPI CSI - 2*4-lane/4*2-lane/4+2*2-lane
Display	- 1 × HDMI 2.0 TX, supporting up to 4K@60fps - 2 × 4-lane MIPI DSI: single channel supports up to 1920×1200@60fps; dual channels support up to 2560×1600@60fps and 3840×2160@45fps. - 2 × RGB, supporting up to 1920×1080@60fps - 4 × LVDS: single channel supports up to 1366×768@60fps; dual channels support up to 1920×1080@60fps - 1 × eDP 1.3, supporting up to 2.5K@60fps and 4K@30fps - Supports dual display
Audio	- Built-in audio codec, supporting 3-channel audio output and 3-channel audio input - 4 × I2S/PCM, supporting up to 16 channels, sampling rate from 8 kHz to 384 kHz, bit depth from 8 bit to 32 bit - OWA - DMIC
Other Peripherals	· 1 x USB2.0 Host, 1 x USB2.0 DRD, 1 x USB3.1 DRD&PCIe2.1 Combo · 2 x GMAC (10/100/1000 Mbps port with RGMII and RMII interfaces) · 2 x CAN· 10 x UART · 8 x I2C (TWI)· 4 x SPI· 2 x IR RX, 1 x IR TX · 30-ch PWM, 21-ch GPADC, 2-ch LRADC · SDIO 3.0, SPIF (Octal I/O)
RAM	LPDDR4X: Optional 2GB/4GB, can be customized and replaced in bulk.
ROM	eMMC: Optional 32GB/64GB, customizable and replaceable in bulk.

PMIC	Special high-efficiency integrated power management chip, the official recommended power solution by Allwinner.
Connector	4 anti-reverse BTB connectors with 0.5mm pitch, 4×80P, totaling 320 pins.

The resource parameters of the development board are shown in the table below.

Table 2 Development Board Resource Parameters

ATK-DLT527B Development Board Resource Parameters	
Core board	ATK-CLT527B Core Board, Allwinner T527 Processor
	RAM: LPDDR4X 2GB/4GB optional
	ROM: EMMC 32GB/64GB optional
LED Indicator	Power indicator * 1
	System indicator
	4G/5G module status indicator * 1
Function Keys	RESET key PWRON key UPDATE key, used to enter download mode User key * 6
Wired Communication	Gigabit Ethernet port * 2
	RS485 * 2
	CAN * 2
	USB2.0 HOST * 2
	USB3.0 HOST * 1, defaults to act as the master; can be configured as the slave via software.
	USB 2.0 OTG * 1, equipped with a USB Type-C connector, supporting firmware burning and ADB debugging.
	Debug UART * 1, using a USB Type-C connector
	JTAG interface * 1
	I2C interface * 4
	TTL UART interface * 4
	SPI interface * 2
Wireless Communication	WiFi & BT antenna interfaces * 2
	WiFi & BT Module (RTL8733BU) * 1
	Mini PCIe 4G/5G module interface * 1
	Nano SIM card slot * 1
External Storage	TF card slot * 1
Audio Function	Headphone jack * 1, supports 3.5mm 4-pole headphones, supports audio playback and recording
	Microphones * 2, dual-channel recording
	Speaker (on the back of the baseboard) * 1
	Speaker interfaces * 2, capable of connecting two external speakers for dual-channel audio playback
Video Input	MIPI CSI port * 2

Screen Display (Dual Display Supported)	MIPI DSI interface * 2, support Dual Link
	LVDS interface * 1
	Parallel RGB interface * 1
	HDMI display interface * 1
	DP display interface * 1
Real-Time Clock (RTC)	External RTC real-time clock chip * 1
Analog Input	Adjustable potentiometer, resistance 10KΩ * 1
	ADC input channels * 19
Expansion IO	3.3V Expansion IO * 10
	1.8V Expansion IO * 22
Power Related	DC-12V input interface * 1
	5V output interface * 3
	3.3V output interface * 3
	Power toggle switch * 1
	RTC battery holder * 1, with CR1220 coin cell battery
Others	Fan interface * 1, which can be connected to a cooling fan and is powered by 12V.

2.3 Software Parameters

The development board supports running different operating systems and various drivers, as shown in the table below.

Table 3 Software Parameters

System and Driver Support		
Operating System	Linux Buildroot (Default factory system, recommended)	
	Android13	
	Debian (Version included with Allwinner SDK, unsupported)	
GUI	QT 5.15.8	
	LVGL 8.3.2	
Driver Support	PMIC	eMMC
	LPDDR4X	ISP
	ADC	GPU
	LED	RTC
	KEY	I2C
	CAN	SPI
	I2S	UART
	RS485	USB2.0 HOST
	USB2.0 OTG	HDMI OUT
	USB3.0 HOST	eDP
	AUDIO OUT	MIC IN
	PWM	SD Card
	4G	5G RedCap

	WiFi	Bluetooth
	MIPI camera	Touch screen
	MIPI screen	LVDS screen
	FAN	Ethernet
Remarks	Provides Tina5.0 Linux and Android 13 SDK source code packages for easy use and supports secondary development.	

Chapter 3. Electrical Characteristics

The electrical characteristic parameters of the development board are shown in the following table.

Operating Parameters	Minimum Value	Typical Value	Maximum Value
Operating Voltage of Development Board	/	12.0V	13.0V
Operating Voltage of Core Board	/	4V	/
Operating Temperature of Development Board (Commercial Grade)	0℃	/	+70℃
Operating Temperature of Core Board (Industrial Grade)	-40℃	/	+85℃

Note: Some chips on the baseboard of the development board are commercial grade with an operating temperature range of 0℃ ~ +70℃. It is recommended to operate within this temperature range to avoid affecting the service life of the board.

The reference power consumption values of the development board are shown in the following table.

No.	Test Case	Typical Voltage	Typical Current	Typical Power Consumption
1	Development board powered on, no system running.	12V	80mA	0.96W
2	The development board only runs the factory-default system without connecting any external modules.	12V	205mA	2.46W
3	The development board is connected to an ALIENTEK 5.5-inch MIPI screen with a resolution of 720×1280, running the factory-default QT interface.	12V	320mA	3.84W

Note: The development board has multiple interfaces. The actual power consumption is related to the number of peripherals used, external modules connected, and other factors. It needs to be measured according to actual usage scenarios; the data provided here is for reference only.

Chapter 4. Mechanical Structure

4.1 PCB Parameters

Parameters	Core board	Development board
PCB Size	60mm*50mm	180mm*140mm
PCB Thickness	1.6mm	1.6mm
PCB Layer Count	10 layers	4 layers
PCB Process	Immersion Gold	Immersion Gold
Number of Mounting & Fixing Holes	4	-

4.2 Core Board Mechanical Dimensions

The dimensions of the ALIENTEK ATK-CLT527B core board are 60mm × 50mm. The mechanical drawing is shown in the figure below.

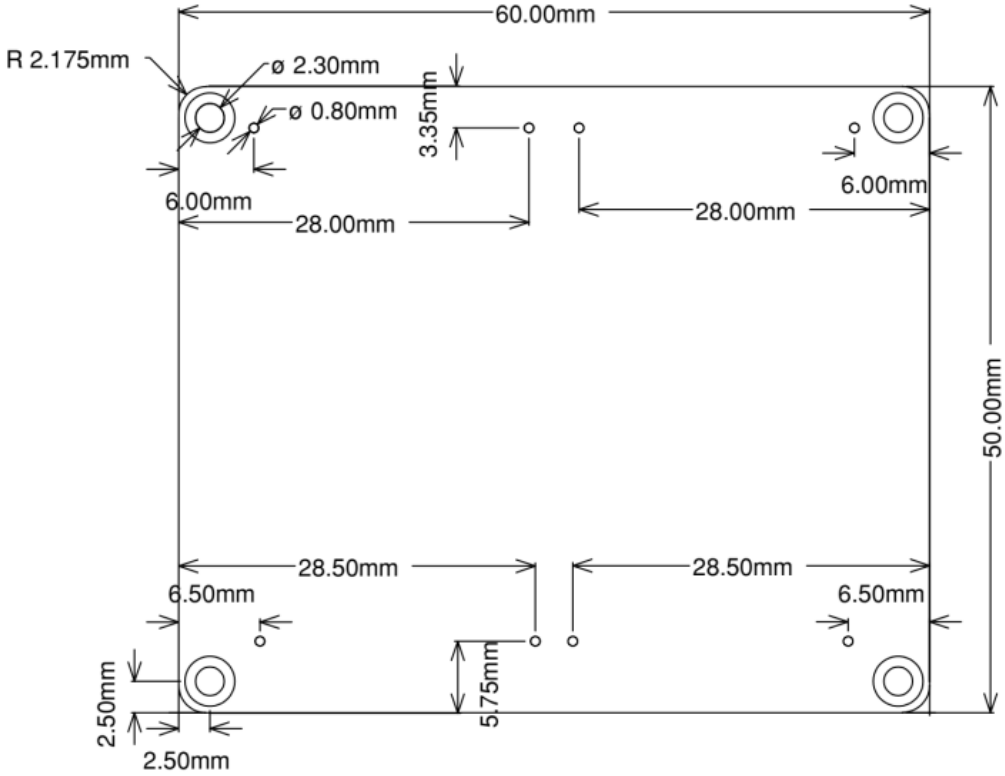


Figure 4.2-1 Core Board Mechanical Drawing

Note: The original file path of the core board mechanical drawing in the above figure is:

Development Board Network Disk Drive A - Basic Documents → 03. Core Board Documents → 03. Core Board Mechanical Structure → ATK-CLT527B V1.0 Mechanical Drawing.pdf

4.3 Development Board Mechanical Dimensions

The dimensions of the ALIENTEK ATK-DLT527B development board are 180mm × 140mm. The mechanical drawing is shown in the figure below.

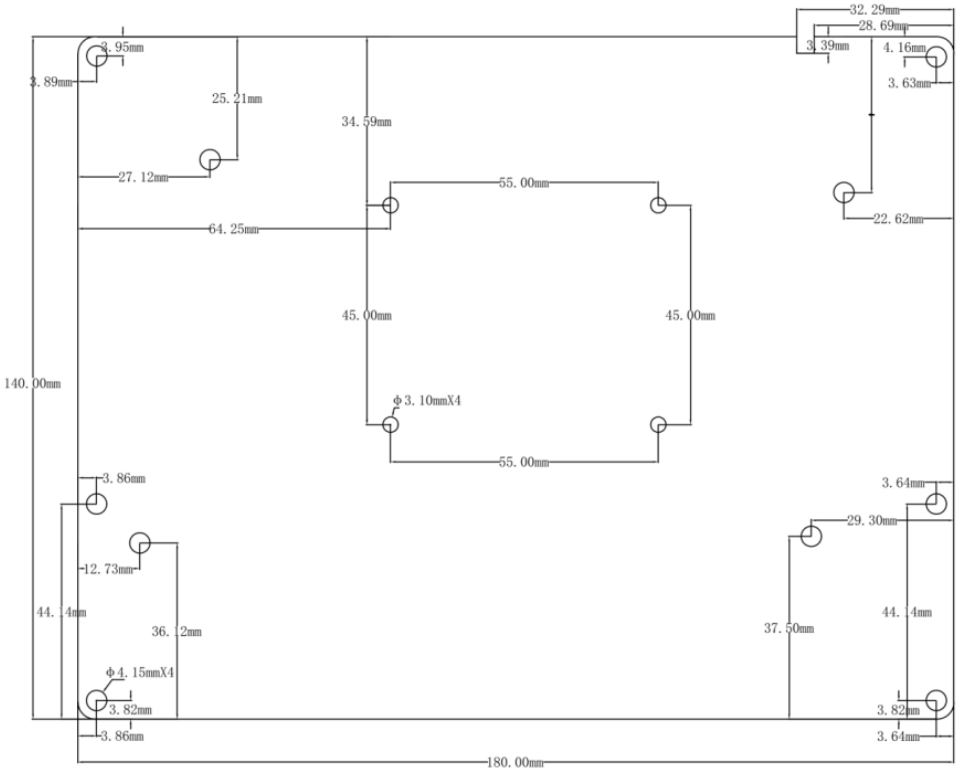


Figure 4.3-1 Development Board Mechanical Drawing

Note: The original file path of the development board mechanical drawing in the above figure is: Development Board Network Disk A - Basic Documents → 07. Hardware Documents → 04. Development Board Mechanical Dimensions → ATK-DLT527B V1.0 Mechanical Drawing.pdf

Chapter 5. Development Board Resources

ALIENTEK provides abundant software resources for the ATK-DLT527B development board, including but not limited to Tina5.0 Linux SDK, Android 13 SDK, Linux Buildroot & Android 13 system firmware, Qt source code, relevant development tools and development environments, etc. Corresponding development documents are also provided, covering development environment setup, SDK compilation, packaging & burning guidance, hardware testing guidance, hardware design references, and more.

In addition, ALIENTEK has specially compiled a series of materials for core board users that will be used during the development phase, involving baseboard schematics, baseboard design references, mechanical structures, component packages, connector specifications, Linux Buildroot & Android system firmware and source code, compilers, software packages, etc., to facilitate rapid development and improve development efficiency for users.

All supporting resources for the development board are available for download via cloud storage.

5.1 Resource Download

ATK-DLT527B Development Board Resource Download Center:
http://www.openedv.com/docs/boards/arm-linux/T527%20Linux.html

5.2 Documentation Description

Level-1 Directory of Chinese Documents in NetDisk A of the Development Board:

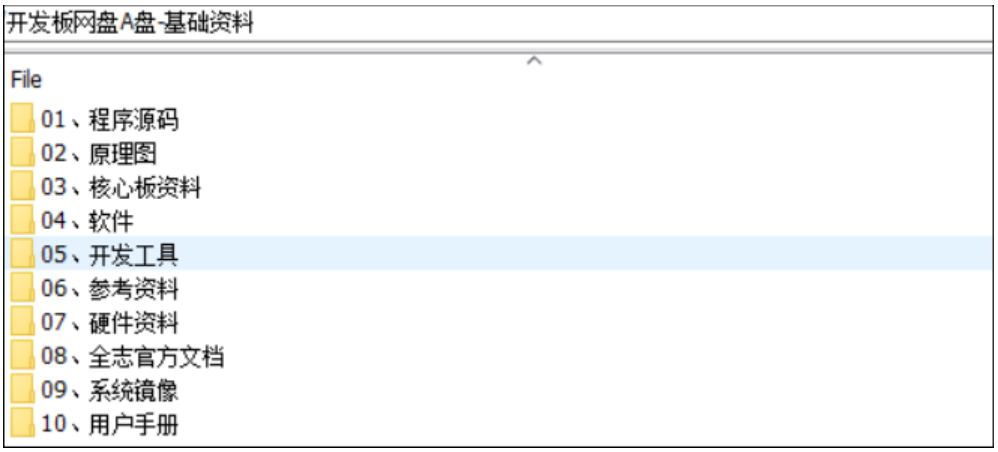


Figure 5.2-1 Level-1 Directory of Documents on Disk A of the Development Board

Description of the Document Directory on Disk A of the Development Board:

Directory	Description
01_codes	Source code collection, such as the source code of comprehensive Qt routines
02_sch	Schematic Diagram of the Development Board Baseboard
03_core_board_data	Includes core board related materials: core board pin definitions, documentation of on-board chips of the core board, mechanical structure of the core board, baseboard design reference materials, specifications, certification certificates, package libraries, etc.

04_softwares	Collection of required software tools, including serial port terminal software, FTP client software, USB-to-serial driver, VMware virtual machine software, etc.
05_tools	Collection of development tools, including cross-compiler, ADB debugging tool, programming tool, Allwinner official development tools, etc.
06_reference_data	Reference materials collection, including protocol manuals, ARM manuals and other reference documents.
07_hardware	Collection of hardware-related documents, including documentation of on-board chips of the baseboard, development board specifications, development board package libraries, mechanical dimensions of the development board, certification certificates, etc.
08_allwinner_reference	Includes development documents provided by Allwinner official, such as hardware documents, software files, tool files, etc.
09_system_image	Includes Linux Buildroot system firmware and Android 13 system firmware. The development board comes pre-installed with the Linux Buildroot system at the factory.
10_user_manual	Stores documents provided to users by ALIENTEK. However, this directory is empty and has been moved to the Yuque online documentation.

Note: Due to limited space, the corresponding directory contents are not elaborated here. Please download the development board materials for detailed reference.

The level-1 directory of the development board's NetDisk B is shown below:

Figure 5.2-2 Level-1 Directory of Development Board Disk B Materials

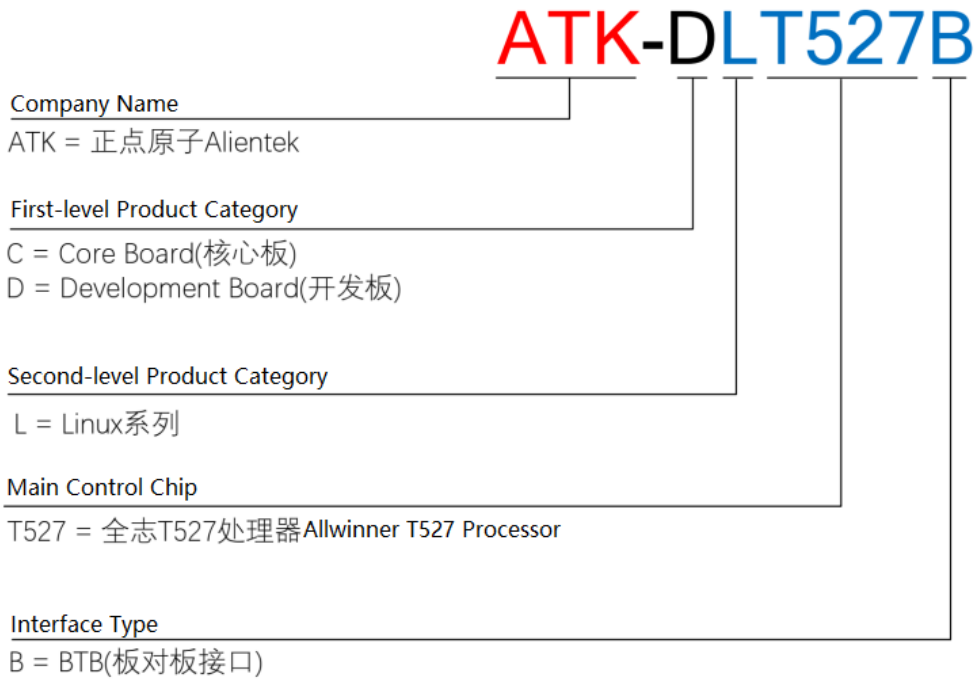
Description of Development Board Disk B Material Directory:

Contents	Explanation
01_Ubuntu Development Environment	To facilitate rapid development for customers, our company provides an Ubuntu virtual machine image with a pre-configured basic development environment. Customers can directly use this Ubuntu virtual machine for software development on the T527 platform, without the need to build an Ubuntu development environment from scratch.
02_ATK-DLT527 development board SDK	Supporting SDKs include Tina5.0 Linux SDK and Android13 SDK, allowing users to conduct secondary development based on these SDKs.

Chapter 6. Development Board Model

Development Board Model: ATK-DLT527B.

The model parameters are explained as follows.



Chapter 7. Development Board Compatible Modules

The ATK-DLT527B development board is equipped with abundant peripheral interfaces for users to conduct functional tests and verifications. Users can purchase corresponding accessories as needed for testing, including the OV13850 camera module, RGB screen, LVDS screen, 4G module, etc.

Chapter 8. Precautions and maintenance

Notes

- Do not plug and unplug peripheral modules with power!
- Before using the product, please carefully read this manual and related development manuals, and pay attention to the applicable matters of the platform.
- Follow all instructions and warnings on the product.
- Please use this product in a cool, dry and clean place.
- Please keep the product dry. If any liquid splashes or soaks, power off immediately and let dry thoroughly.
- Do not use organic solvents or corrosive liquids to clean the product.
- Do not use or store this product in dusty, dirty and messy environment.
- If not used for a long time, please package this product, pay attention to moisture-proof and dust-proof.
- Pay attention to the ventilation and heat dissipation of the product during use to avoid component damage caused by excessive temperature during operation.
- Do not use this product in alternating hot and cold environment to avoid dew damage to components.
- Do not treat this product roughly, drop, knock or shake violently may damage the line and components.
- Pay attention to anti-static when using this product.
- FPC flexible cable is fragile, when plugging cable, pay attention to check whether the metal at both ends of the cable is misplaced and falling off.
- All products have passed the product test before shipment. Please use the development board corresponding to the ALIENTEK for power on test for the first time.
- Do not repair or disassemble the company's products by yourself. If the product fails, please contact the company in time for maintenance.
- Unauthorized modification or use of unauthorized parts may damage the product, the resulting damage will not be repaired.

Chapter 9. After sales service

9.1 Terms of after-sales service

1). After receiving the goods, please open them in front of the express, and sign after acceptance. If you find that the goods are less after signing, take photos in time and contact the seller's customer service to explain the situation within 15 days. If the feedback is lack of goods after 15 days, we will not reissue the goods. Other reasons notwithstanding).

2). 15 days -1 month: we are responsible for the return freight repair of product problems. Human factors damage expensive main chip or LCD screen, touch screen. The buyer needs to pay the cost and one time shipping fee, no maintenance fee.

3). 1-3 months: the problem of the product itself (non-human factors), we are responsible for the delivery of the past freight maintenance. If the main chip is burned out and the LCD screen and touch screen are damaged, the buyer needs to pay the cost, and the maintenance fee is not charged.

4) After 3 months: the buyer shall bear the return freight and the cost of chip, LCD screen and touch screen. No service charge.

9.2 After-sales Support

Technical support:

QQ group: ALIENTEK Allwinner Communication group: 1032186104

ALIENTEK Allwinner User Group: 2152038686 (order number required)

Taobao shop: ALIENTEK flagship store

Tel: 020-38271790 (Linux technical support via reception)

Forum: http://www.openedv.com/forum.php